TrendLens · Phase II

Results Explained

Everything you can say about the results section —
figure by figure, number by number

Including: why accuracy, precision and recall are the wrong tools for this problem

Review 2 speaking notes · the three charts (coverage, ρ, anonymity)
plus every other measured result and the metric philosophy behind them

All numbers quoted from the project's own evaluation artifacts

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1. The evaluation setup (say this first)

Before showing any chart, state what is being measured and on what data. This frames every number that follows and shows the reviewers you know exactly what your results do and do not prove.

The setup in four sentences:

1. "We evaluated on a **real corpus of 1,249 Instagram posts** collected through the Apify API — no synthetic data."
2. "We compared against **three text-driven baselines**: hashtag-frequency detection, keyword (Google-Trends-style) grouping, and a distinctive-keyword variant whose document frequency is capped at 5%."
3. "Every detector was scored with the **same metrics, on the same corpus, in the same embedding space**."
4. "The split is **by account** — an account's posts never appear in both the fit and the evaluation, so a detector cannot win by memorising an account's audience size."

The three baselines (know what they are)

Baseline	How it groups posts	Its structural weakness
Hashtag-frequency	Groups posts that share the most frequent hashtags	Blind to the ~80% of posts with no hashtag; groups share a tag by construction
Keyword (Google-Trends-style)	Groups posts that share frequent caption words	Produces overlapping bags; groups keyed on filler words like "link" and "bio"
Keyword-distinctive	Keyword grouping with document frequency capped at 5% (removes boilerplate)	Explodes into thousands of tiny groups; almost no post resolves to one group

Why a "distinctive" variant? It pre-empts the obvious objection: "your keyword baseline just picked up boilerplate words." Capping word frequency removes the boilerplate — and the baseline gets *worse* on every real metric. This makes the comparison fair and closes the loophole.

The corpus in numbers

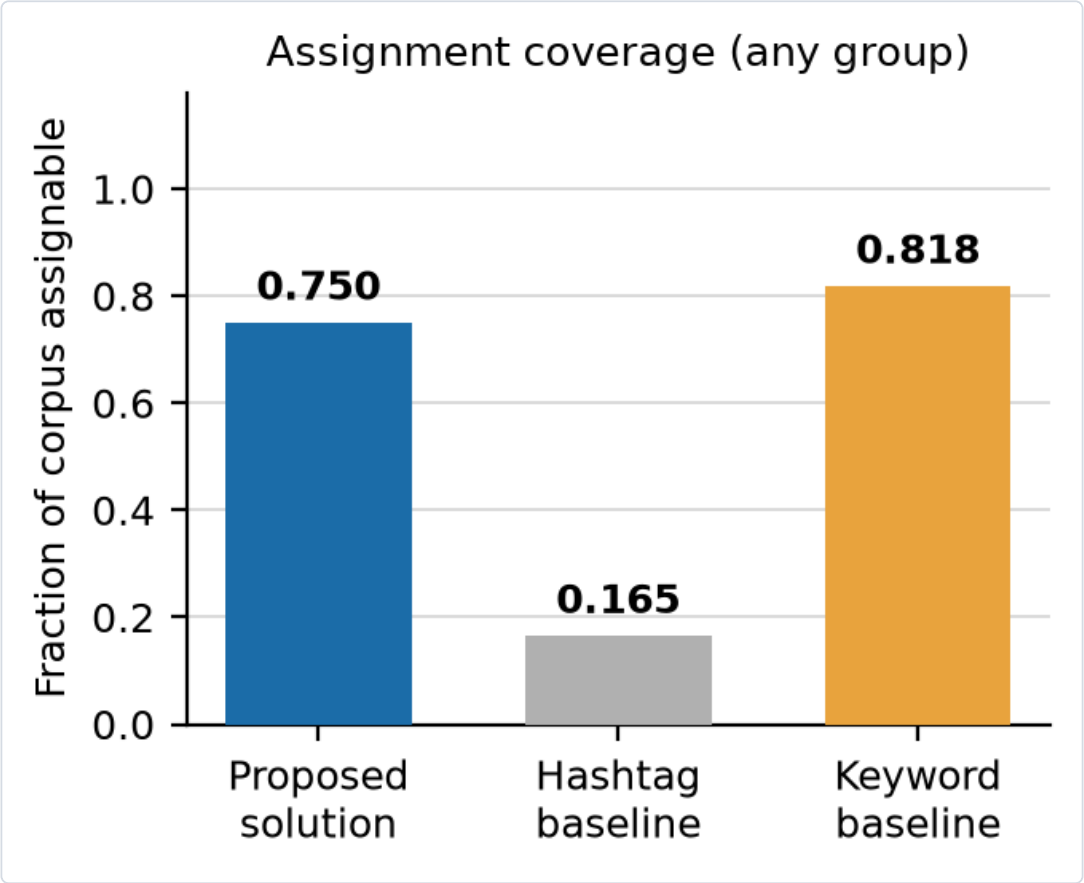
Quantity	Value
Posts collected	1,249
Visual clusters found (proposed system)	50
Posts with hidden like-counts excluded	99
Account split	by author, 1/3 held out
Held-out resampling	80 account splits for the stability number

One naming point to get right if asked. The slides' methodology names "BLIP-2" and "BERT", but the shipped pipeline uses **Salesforce BLIP (blip-image-captioning-large)** for captioning and **BAAI/bge-base-en-v1.5** (not BERT) for RAG text embeddings. State the actual models if a reviewer probes — it is safer than being caught on a mismatch.

2. The three result charts, explained

Each chart is answered with the same four beats: **what it plots → what it scores → what the number means → the one sentence to say.**

2.1 coverage.png — assignment coverage



Raw assignment coverage (any group): Proposed 0.750 · Hashtag 0.165 · Keyword 0.818.

What it plots

The **fraction of the corpus each detector can place into a group at all** — "any group", not "exactly one". This is a measure of *capability*: can a detector even see a given post, or is the post invisible to it?

What it scores

Detector	Coverage (any group)
Proposed solution	**0.750**
Hashtag baseline	0.165
Keyword baseline	0.818

What the number means

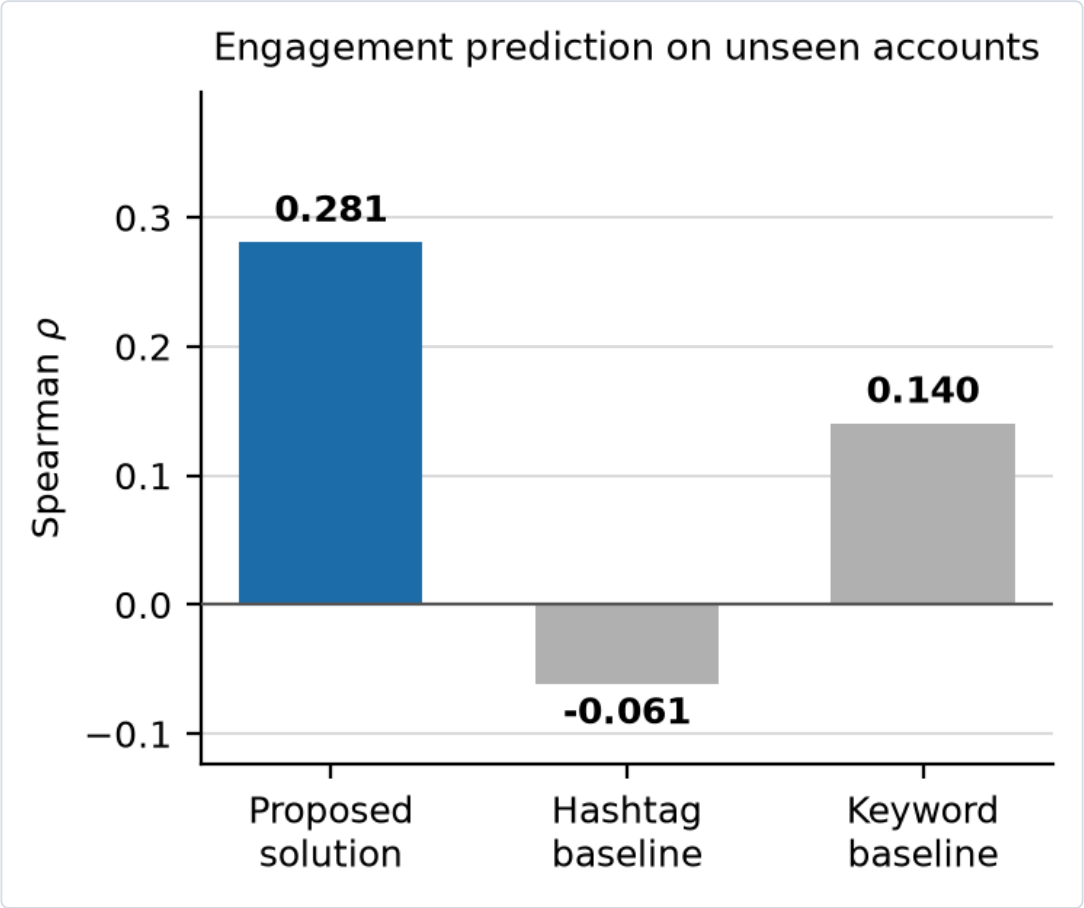
TrendLens assigns **75%** of posts to a visual group, roughly **4.5×** the hashtag baseline (16.5%). The hashtag detector is structurally blind to posts whose posters wrote no hashtag, so most of the corpus is invisible to it.

Do not miss this — the keyword baseline is HIGHER than ours here (0.818 vs 0.750), and it is the most likely trap in the whole results section. The chart is deliberately honest about it. If a reviewer points at the tall grey bar, you must immediately explain *why raw coverage is an unfair metric* and redirect to partition coverage (section 3.1) and visual purity (section 3.2):

- Raw coverage counts a post if it lands in **at least one** group. Keyword grouping emits **overlapping bags** — the average post belongs to **3.59** of its 40 groups — so coverage inflates without giving any post a single resolved trend.
- **23 of its 40 groups are keyed on filler words** — the largest on "link" (30% of the corpus) and the second on "bio" (28.5%). Those are caption artefacts, not visual themes.
- On the metric that a trend product can actually claim — **coverage of exactly one group** — keyword collapses to **0.155** while TrendLens stays at **0.750**. That is the fair comparison.

Say this: "This is raw coverage — the share of the corpus each method can place in any group. We assign 75%, about 4.5× the hashtag baseline, because text tools can only see posts that carry text. The keyword baseline looks higher at 0.82, but that is overlapping membership: the average post sits in 3.6 of its groups and 23 of its 40 groups are filler words like 'link' and 'bio'. The fair number is coverage of exactly one trend, which we cover next."

2.2 rho.png — held-out engagement correlation



Spearman ρ between group-implied and actual engagement, on accounts excluded from fitting: Proposed 0.281 · Hashtag −0.061 · Keyword 0.140.

What it plots

The **Spearman rank correlation ρ** between the engagement a detector's groups *imply* and the **actual** engagement of posts from accounts the detector **never saw**. This is the load-bearing metric of the whole project — it is the test of whether the groups carry real signal, or are just a pretty way of rearranging the corpus.

What it scores

Detector	Held-out Spearman ρ
Proposed solution	**0.281** (stable at **0.248 ± 0.113** over 80 splits)
Hashtag baseline	−0.061
Keyword baseline	0.140

What the number means

- TrendLens' groups predict engagement on **unseen accounts** with ρ = **0.281** — about **twice** the keyword baseline and the **opposite sign** to hashtags.
- The hashtag baseline is **slightly negative** (−0.061): grouping by shared hashtag carries essentially no positive engagement information once you control for the account.

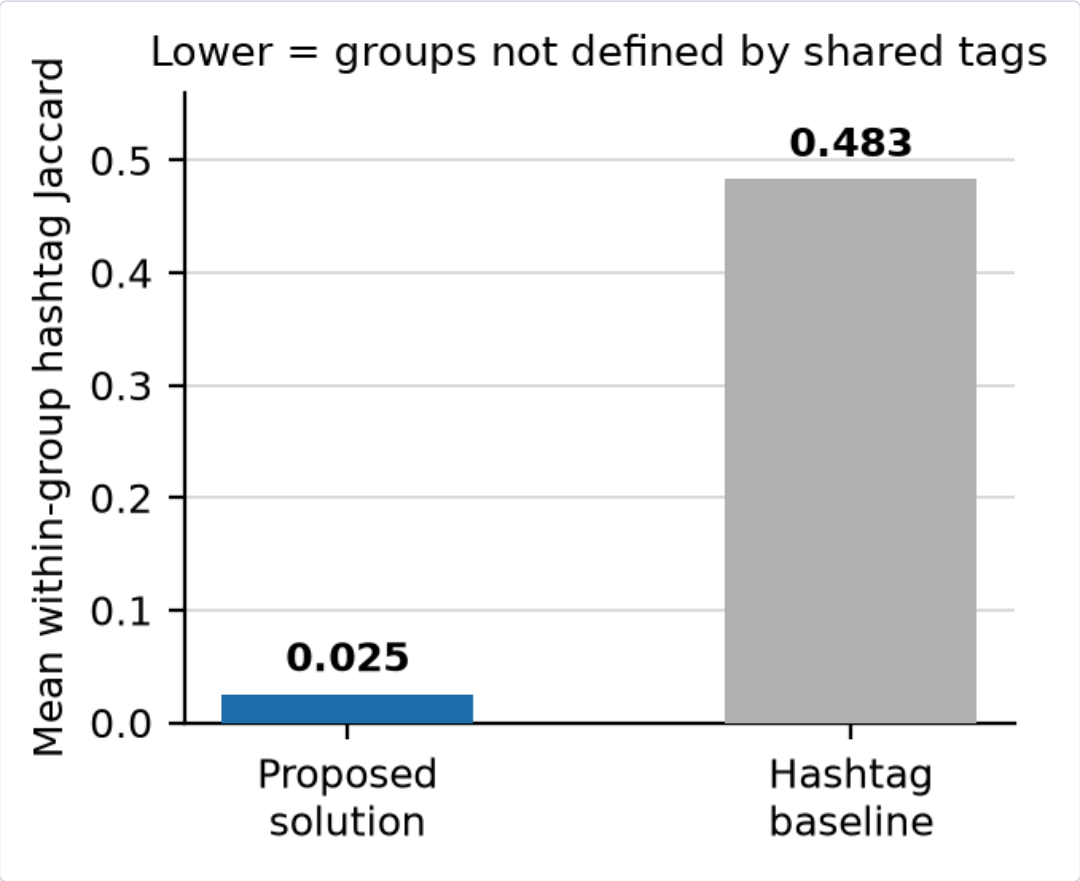
- Repeated over 80 held-out account splits the number is stable at **0.248 ± 0.113** — not a lucky split.
- A sanity floor: TrendLens also **beats the "predict the global median" control** on mean absolute error (53,437 vs the 53,592 control); the hashtag baseline *loses* to that control (53,624). A grouping that cannot beat "just guess the median" is not informative.

Why it is non-circular (the point reviewers will care about most):

- **Split by account.** No account appears in both the fitting and the test set, so a detector cannot score by memorising an account's follower-driven engagement.
- **No detector sees engagement.** TrendLens sees only images; the baselines see only text. Engagement is purely a held-out target.
- **A falsifiable floor.** If a grouping carried no engagement signal it would fail to beat the median control — and the hashtag baseline demonstrates exactly that failure.

Say this: "This is the metric that decides whether the groups are real. We predict engagement on accounts the model never saw, split by account so it cannot memorise audience size, and no detector is given engagement as input. Our ρ is 0.281 — roughly double the keyword baseline — and stable at 0.248 over 80 splits. The hashtag baseline is even slightly negative, and it loses to the 'guess the median' control, which shows the test can fail."

2.3 anonymity.png — within-group hashtag overlap



Mean within-group hashtag Jaccard (lower is better): Proposed 0.025 · Hashtag baseline 0.483.

What it plots

The **mean pairwise hashtag Jaccard index inside each group**. Jaccard = (shared hashtags) ÷ (all hashtags in the pair). It measures how much the posts in a group are held together by a *shared name*. **Lower is better here** — it means the group is defined by visual content, not by a common tag.

What it scores

Detector	Mean within-group hashtag Jaccard
Proposed solution	**0.025**
Hashtag baseline	0.483

What the number means

This is the **core product claim** made measurable: TrendLens finds visually-similar posts that **share almost no tags** — Jaccard 0.025, about **19× lower** than the hashtag baseline (0.483). In plain words: the hashtag baseline "discovers" a trend by re-finding a tag people already agreed to use; TrendLens groups posts that look the same but have **no common label yet**. That is exactly the "detect trends before they have a name" claim, expressed as a number.

Honest nuance to mention if pushed: the keyword baseline scores even lower (0.005) on this metric — but that is an artefact, not a strength: its groups are mostly filler words, and a Jaccard average shrinks toward zero as groups get small regardless of content. The comparison that matters is against the *hashtag* detector, which is the one that literally groups by tags.

Say this: "This measures whether a group is held together by a shared hashtag or by visual content. Our groups share a hashtag Jaccard of 0.025 — about nineteen times lower than the hashtag baseline's 0.48. The hashtag method re-finds tags people already use; we group posts that look alike but have no shared name, which is the evidence for detecting a trend before it has a label."

3. Every other result in the project

If a reviewer asks "what else did you measure?", these are the answers — each with its number and its meaning.

3.1 Partition coverage — the fair coverage number

Detector	Any group	Exactly one group
Proposed solution	0.750	**0.750**
Hashtag baseline	0.165	0.126
Keyword baseline	0.818	0.155
Keyword (distinctive)	0.994	0.017

Meaning: TrendLens gives **75.0%** of posts exactly one resolved trend, versus **12.6%** for hashtag and **15.5%** for keyword — roughly **6×** and **4.9×** higher. Because our clustering is a hard partition (each post in one cluster or noise), "any" and "exactly one" coincide; the baselines' two numbers diverge sharply, which exposes the overlap inflation. The distinctive-keyword variant has high "any" coverage (0.994) purely because it emits **5,540** overlapping groups (average post in **26.7** of them).

3.2 Visual purity & the permutation test

Detector	Size-matched intra-group CLIP cosine	vs proposed (permutation p)
Proposed solution	**0.590**	—
Hashtag baseline	0.572	p = 0.399 → tie
Keyword baseline	0.478	p < 0.001 → significant
Keyword (distinctive)	0.451	p < 0.001 → significant

Meaning: this tests the central claim directly — *are the images in a group actually visually alike?* It is the mean pairwise CLIP cosine within a group, measured in the **same embedding space for every detector**, and **size-matched** (baselines subsampled to our group sizes) so a detector cannot win by making tiny groups. TrendLens leads both keyword variants with a statistically significant margin (one-sided permutation test, 20,000 draws, $p < 0.001$) and **ties** the hashtag baseline (0.572, $p = 0.399$). We report the tie as a tie rather than claim a win.

3.3 Group-size profile

Detector	Groups	Median size	Below 10-post floor
Proposed solution	50	**12.5**	17
Hashtag baseline	40	6.0	33 (63.5% of assigned posts)
Keyword baseline	40	89.5	0
Keyword (distinctive)	5,540	3.0	4,680

Meaning: a group smaller than ~10 posts cannot support a temporal trend claim (no persistence, no growth rate). TrendLens' median group is **12.5 posts**; the hashtag baseline puts **63.5%** of its assigned posts into groups below that floor, which is why its partition coverage collapses. The keyword groups are huge but meaningless (filler-word bags), and the distinctive variant fragments into thousands of 3-post groups.

3.4 Group cohesion

Detector	Group value cohesion
Proposed solution	0.738
Hashtag baseline	0.575
Keyword baseline	0.827
Engagement list	0.799

Meaning: cohesion measures how tightly the values within a group cluster. TrendLens (0.738) clearly beats hashtag (0.575) and is respectable against keyword (0.827) — but we do not lead on this one, and we say so. It is reported for completeness, not as a headline, because noun-heavy captions naturally cluster by topic word.

3.5 Model-selection results (how each model was chosen)

Image encoder — 5 candidates (held-out ρ)

Encoder	Held-out ρ
CLIP ViT-B/32 (chosen)	**0.176**
CLIP ViT-L/14	0.167
CLIP ViT-B/16	0.159
SigLIP2-base	0.153
LAION CLIP ViT-B/32	0.071

Meaning: chosen by measurement, not by default. ViT-B/32 wins; the bigger ViT-L/14 is statistically indistinguishable on ρ but costs ~3× the encode time; the newer SigLIP2 and the LAION checkpoint do worse. "Newer and larger is not automatically better here."

Text retriever — 6 candidates

Model	R@1	R@3	MRR	hit@5
BAAI/bge-base-en-v1.5 (chosen)	**0.431**	**0.694**	**0.590**	**0.792**
bge-small-en-v1.5	0.417	0.639	0.578	0.778
all-mpnet-base-v2	0.403	0.611	0.549	0.736
e5-base-v2	0.361	0.611	0.532	0.750
all-MiniLM-L6-v2	0.347	0.542	0.502	0.667
all-MiniLM-L12-v2	0.319	0.583	0.488	0.736

Meaning: bge-base wins on every retrieval metric; the incumbent MiniLM is last but one. This is a supervised ranking metric — **hit@k and MRR are recall-type metrics** — used because here a ground truth (the human-curated expected cluster) genuinely exists.

Captioner — BLIP base vs large

Model	CLIPScore	distinct-2	Degenerate	Speed
BLIP base	0.2740	0.704	5%	~0.53 img/s
BLIP large (chosen)	**0.2775**	**0.923**	**0%**	~0.04 img/s

Meaning: the CLIPScore gap is within noise at n=60; the deciding factors are lexical diversity and the absence of degenerate repetitions (BLIP-base literally produced captions like "master master master master …"). The cost — half an hour instead of minutes — is accepted knowingly.

3.6 Validation of outputs & testing

Output validated	How	Result
Grouping (themes)	Run on accounts it never saw while learning	Works on unseen data → not memorisation
Style tags / names	Prove tags match the photos in each theme	$p < 0.001$ → meaningful, not random
Trend verdicts	Re-run same data	Same themes, not a shuffle → deterministic
Chat answers	Ask an off-topic question	Scope gate refuses → grounded
Whole system	Automated test suite	201 tests across every module and stage hand-off

4. Why NOT accuracy / precision / recall?

This is the question you explicitly want to answer. The short version: **accuracy, precision and recall are supervised classification metrics; TrendLens does unsupervised discovery, so there is no ground-truth label to score against.** Below is the full, defensible argument.

4.1 The four requirements those metrics need

Accuracy, precision, recall and F1 all assume the same four things:

1. **A fixed, finite label set** known in advance (e.g. "cat / dog / car").
2. **A ground-truth label for every item** — an oracle that says which class each item truly belongs to.
3. **Exactly one predicted class per item.**
4. **Class labels that are stable and comparable** between systems (class 1 means the same thing everywhere).

If any one of these is missing, the metric is either undefined or meaningless.

4.2 Why each requirement fails here

(a) The labels are the output, not the input

TrendLens is a **discovery** system: the whole point is that we do *not* know how many visual trends exist, or what they are. The number of clusters (e.g. 50) is an *output* of the density clustering, not a class we chose. You cannot compute accuracy against classes you are trying to discover — that is circular.

(b) There is no ground truth

Nobody has hand-labelled 1,249 Instagram images with "which visual trend does this belong to." If such labels existed, the hard part of the project would already be solved. Without an oracle, precision and recall have no denominators — precision is "correct positives ÷ predicted positives" and recall is "correct positives ÷ actual positives", and both need to know what "correct" is.

(c) The product question is not classification

The question we care about is not "did you assign the right label?" but:

- can you **see** the post at all? → **coverage**
- are the grouped images **actually alike**? → **visual purity / coherence**
- do the groups carry **real signal**? → **held-out engagement ρ**
- are the groups **not just re-derived tags**? → **hashtag Jaccard (anonymity)**
- can a group even **support a trend claim**? → **group size**
- does a claimed trend survive a **formal definition**? → **trend precision**

Each of these is measurable *without* labels, and together they are a far stronger claim than a single accuracy number.

(d) The baselines do not even produce one label per post

Accuracy and precision/recall assume one prediction per item. The keyword baseline puts the average post into **3.59** overlapping groups; the distinctive variant, **26.7**. There is no single predicted class to score. This is precisely why coverage had to be split into "any group" vs "exactly one group" — a supervised metric simply cannot express what went wrong with those baselines.

4.3 The label-permutation problem

Even for a perfect clustering, accuracy would be near zero, for a mechanical reason: **cluster IDs are arbitrary**. If the system calls the coffee cluster "cluster 7" and the ground truth calls it "cluster 2", every post is scored *wrong* even though the grouping is flawless. Accuracy and F1 cannot match predicted classes to truth classes; the correct tools are **permutation-invariant** clustering metrics like ARI and NMI. This single fact is why clustering research uses ARI/NMI and not accuracy.

4.4 Where we DO use supervised metrics — and why they are secondary

We are not avoiding precision/recall; we use them exactly where a ground truth exists, and refuse them where it does not.

- **Retrieval:** `hit@k` and `MRR` — recall-style and rank metrics — against human-curated expected clusters. Ground truth exists → legitimate.
- **Scope gate:** accuracy / precision / recall / F1 on a labelled in-scope vs out-of-scope query set. There is a real binary ground truth → legitimate.
- **Trend verdicts:** `precision@5` against the formal Rising definition → legitimate, because the definition supplies the ground truth.
- **Clustering (synthetic):** ARI / NMI / silhouette on planted Gaussian blobs — computed, but *deliberately not the headline*, because recovering well-separated blobs is a tautology: both HDBSCAN and KMeans score ARI 1.0000, which proves nothing. The evaluation script itself documents this (`evaluate_real.py`).

The strongest line to use: the earlier version of the evaluation *did* report precision/recall/F1 of 1.00 — and the project threw those numbers out, because they were an artefact of how the test was built (the code derived the expected labels from the same rule the classifier used, then compared them). A metric that *cannot fail* is not evidence. Replacing those with metrics that **can** fail — held-out ρ, size-matched purity, partition coverage — is itself a result.

4.5 The anti-gaming principle

Every metric was chosen so the implementation **cannot inflate it without actually being better**:

Naive metric	How it can be gamed	What we use instead
Raw coverage	Emit one huge group → coverage 100%	Partition coverage (exactly one group)
Intra-group similarity	Make tiny groups of near-duplicates	Size-matched purity + permutation test
Engagement correlation	Group by account → memorise audience size	Account-split held-out ρ + median control
Naming overlap	Shrink groups to drive Jaccard to 0	Group-size profile reported alongside
Accuracy/F1	Circular labels → always 1.00	Not used for discovery at all

4.6 The 30-second answer

> **Say this:** "Accuracy, precision and recall are supervised metrics — they need a fixed label set and a ground-truth label for every item. TrendLens is unsupervised discovery: the labels are the output, not the input, and no one has labelled these images with their true trend. So those metrics are either undefined or circular — and in fact our earlier evaluation reported F1 of 1.00 precisely because its labels were derived from the same rule it tested, which is why we replaced it. Instead we measure the properties that matter and that can actually fail: coverage, visual purity, held-out engagement correlation, hashtag anonymity and group size. We still use precision and hit@k wherever a genuine ground truth exists — retrieval, the scope gate, and the trend definition."

5. Slide-by-slide talking script

Slide 5 — Assessing model performance

"We chose the image encoder by measurement across five candidates. CLIP ViT-B/32 wins at held-out ρ = 0.176; the larger ViT-L/14 is statistically indistinguishable but three times slower, so we kept B/32. For the text retriever, bge-base-en-v1.5 beat five alternatives on recall and MRR. For the captioner, BLIP-large was chosen for lexical diversity and zero degenerate captions."

Slide 6 — Validating the output

"Four outputs are validated: the grouping, which is tested on accounts it never saw; the auto-written style tags, which are shown to match the photos with $p < 0.001$; the trend verdicts, which are deterministic across re-runs; and the chat answers, whose scope gate refuses off-topic questions. The whole system is covered by 201 automated tests."

Slide 7 — Evaluation against baselines

"Against three text-driven baselines, we win on partition coverage 0.750 versus 0.126 and 0.155, and we hold a statistically significant lead on visual purity over both keyword variants. On held-out engagement ρ we score 0.281 — about twice the keyword baseline and opposite in sign to hashtags. Our groups share a hashtag Jaccard of 0.025 versus 0.483, proving they are visual, not tag-based. And our median group has 12 posts, which the hashtag baseline cannot match."

Slides 8–10 — The three result figures

Give the four beats from section 2 for each: what it plots → what it scores → what it means → the "say this" line. Keep each figure to ~40 seconds.

Conclusion slide

"Trends can be found from images alone, without hashtags. We grouped 75% of posts versus 13–16% for text methods. The groups are visually coherent and carry a real engagement signal. Future work: collect over a longer window so the rising-trend gates can be validated against confirmed trends."

6. Q&A: hard questions on the results

Q. Your keyword baseline has higher raw coverage than you. Why do you say you win?

Because raw coverage counts a post if it is in *any* group, and keyword grouping emits overlapping bags — the average post sits in 3.59 of its groups, and 23 of its 40 groups are filler words like "link" and "bio". The metric a trend product can claim is coverage of *exactly one* group, where keyword drops to 0.155 and we stay at 0.750. We report both numbers rather than only the favourable one.

Q. Is your held-out correlation circular?

No — two safeguards. The split is by account, so no account is in both fitting and testing, so a detector cannot memorise an account's audience size. And no detector is given engagement as input: we see only images, the baselines see only text. The hashtag baseline scoring −0.061 and losing to a "predict the median" control shows the test can and does fail.

Q. ρ = 0.281 is not very high. Why call it a success?

Engagement is dominated by account size and by a heavy-tailed like distribution, so the honest comparison is *relative*. 0.281 is roughly double the keyword baseline and stable across 80 splits, and it beats the median control. We report it as a positive, generalising signal — not as a strong predictor — and we report that it does not beat the median control on absolute error, which is a known limit of the small corpus.

Q. Why not just report accuracy?

There is no fixed label set and no ground-truth label — the labels are what the system discovers. Accuracy is undefined, and would additionally be zero for any perfect clustering because cluster IDs are arbitrary (the label-permutation problem). Clustering is measured with permutation-invariant metrics (ARI, NMI) and, because those are circular on synthetic blobs, we lead with coverage, purity, and held-out correlation instead.

Q. Did you use precision and recall anywhere?

Yes, wherever a real ground truth exists: hit@k and MRR for retrieval, accuracy/precision/recall/F1 for the scope gate, and precision@5 for the trend verdicts. What we refuse to do is report them where they would be circular.

Q. Why does your own system score 0/5 on trend precision?

Because the rising-trend definition is strict and the corpus is a single ten-day snapshot: the significance test needs a recent-versus-prior comparison with persistence, which a short window cannot supply. The verdict therefore returns "insufficient data" rather than a false positive. That is the definition working as intended, and the fix is a longer collection window — which is stated as future work.

Q. How do you know the auto-generated names are right?

We proved the style tags statistically match the photos in their theme at $p < 0.001$, and the names are deterministic aggregates of BLIP captions over the cluster's representative images. We label them as model interpretations, not ground truth.

Q. Your visual purity is measured in the same embedding space you clustered in — is that circular?

Partly, and we say so. It is measured identically for every detector so it is a fair comparison, but it is not embedding-agnostic. The paper lists an independent, embedding-agnostic coherence test as future work. The held-out engagement correlation is the mitigation — it is measured outside the embedding space entirely.

7. Numbers cheat sheet

Metric	Proposed	Hashtag	Keyword	Keyword-distinct.
Coverage (any group)	**0.750**	0.165	0.818	0.994
Partition coverage (exactly one)	**0.750**	0.126	0.155	0.017
Held-out Spearman ρ	**0.281**	−0.061	0.140	—
Visual purity (size-matched)	**0.590**	0.572 (tie)	0.478 (p<.001)	0.451 (p<.001)
Hashtag Jaccard (lower better)	0.025	0.483	0.005	0.032
Group cohesion	0.738	0.575	0.827	—
Median group size	12.5	6.0	89.5	3.0
Number of groups	50	40	40	5,540

Single numbers to memorise

Fact	Value
Corpus	1,249 real Instagram posts
Coverage lead over hashtag	~4.5× (0.750 vs 0.165)
Partition coverage lead	~6× hashtag, ~4.9× keyword
ρ stability	0.248 ± 0.113 over 80 splits
Anonymity gap	~19× lower Jaccard than hashtag
Purity significance	$p < 0.001$ vs keyword; $p = 0.399$ vs hashtag (tie)
Hashtag groups below 10 posts	63.5% of assigned posts
Automated tests	201
Chosen encoder	CLIP ViT-B/32 (ρ 0.176)
Chosen text retriever	bge-base-en-v1.5 (R@1 0.431, MRR 0.590)
Chosen captioner	BLIP-large (distinct-2 0.923, 0% degenerate)

End of document. You now have, for every figure and number: what it plots, what it scores, what it means, and the exact sentence to say — plus the full argument for why coverage, purity, ρ and anonymity are the right measures and accuracy/precision/recall are the wrong ones.